



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

Introduction to Data Science

Lecture 22: Logistic Regression

Logistic Regression

Supervised Learning

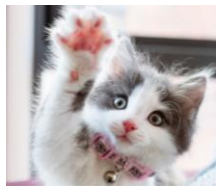
Training data



$y=1$ (cat)



$y=0$ (dog)

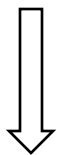


$y=1$ (cat)

...



$y=0$ (dog)



Learning algorithm (optimization involved)

Classifier $h: X \rightarrow \{0,1\}$

For example:

$h \left(\text{image of a dog} \right)$



0

Instead of letting h output $\{0,1\}$, we can consider that given x , h outputs the probability of each class

Main Idea of Logistic Regression

Let's return to the **cat vs. dog** example.

- Logistic regression tries to **learn a formula** that takes the input (pixel values, features, etc.) and **outputs a number between 0 and 1**, representing:
- The **probability** that the input is a **cat** (or a dog)

If the output is close to 1 → probably a **cat**

If close to 0 → probably a **dog**

What is logistic regression model?

- Model the conditional probability of the label for each data

$$P(it\ is\ a\ dog\ | \ ) = ?$$

- Use all labeled samples to estimate the parameters of the conditional probability model.

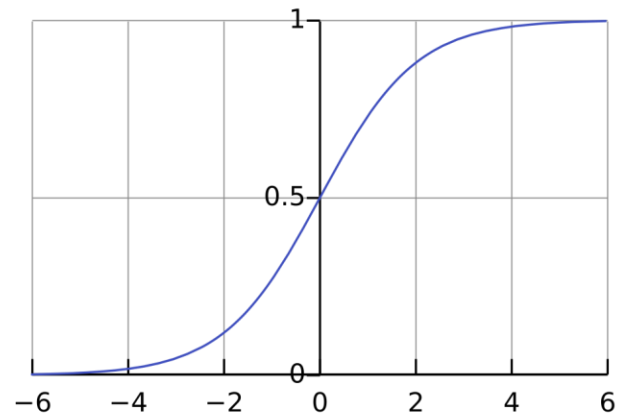
The Logistic (Sigmoid) Function

At the heart of logistic regression is the **logistic function**, also called the **sigmoid function**:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where

- $\sigma(z)$ maps any real number z to a value between **0 and 1**
- This output can be interpreted as a **probability**



How Logistic Regression Does Classification?

To classify a new input x , where x is a numerical representation of an image :



We use the **logistic (sigmoid) function** $\sigma(z)$ to model the **conditional probability** that x belongs to a certain class (e.g., cat or dog):

$$p(it\ is\ a\ dog\ |z) = \sigma(z) \text{ where } z = \theta^T x + b$$

- θ = weights learned from data
- x = feature vector (e.g., image data)
- b = bias term

This means logistic regression **assumes** the probability of an input being a cat (or dog) is based on a **linear combination of features**, passed through the sigmoid function to squash the result between 0 and 1.

What is logistic regression model?

- **Simplest** case (two classes): $y \in \{0, 1\}$
- **Logistic regression model:**

$$p(y = 1|x, \boldsymbol{\theta}, b) = \frac{1}{1 + \exp(-(\boldsymbol{\theta}^\top \mathbf{x} + b))}$$

$$p(y = 0|x, \boldsymbol{\theta}, b) = 1 - p(y = 1|x, \boldsymbol{\theta}, b) = \frac{\exp(-(\boldsymbol{\theta}^\top \mathbf{x} + b))}{1 + \exp(-(\boldsymbol{\theta}^\top \mathbf{x} + b))}$$

What is logistic regression model?

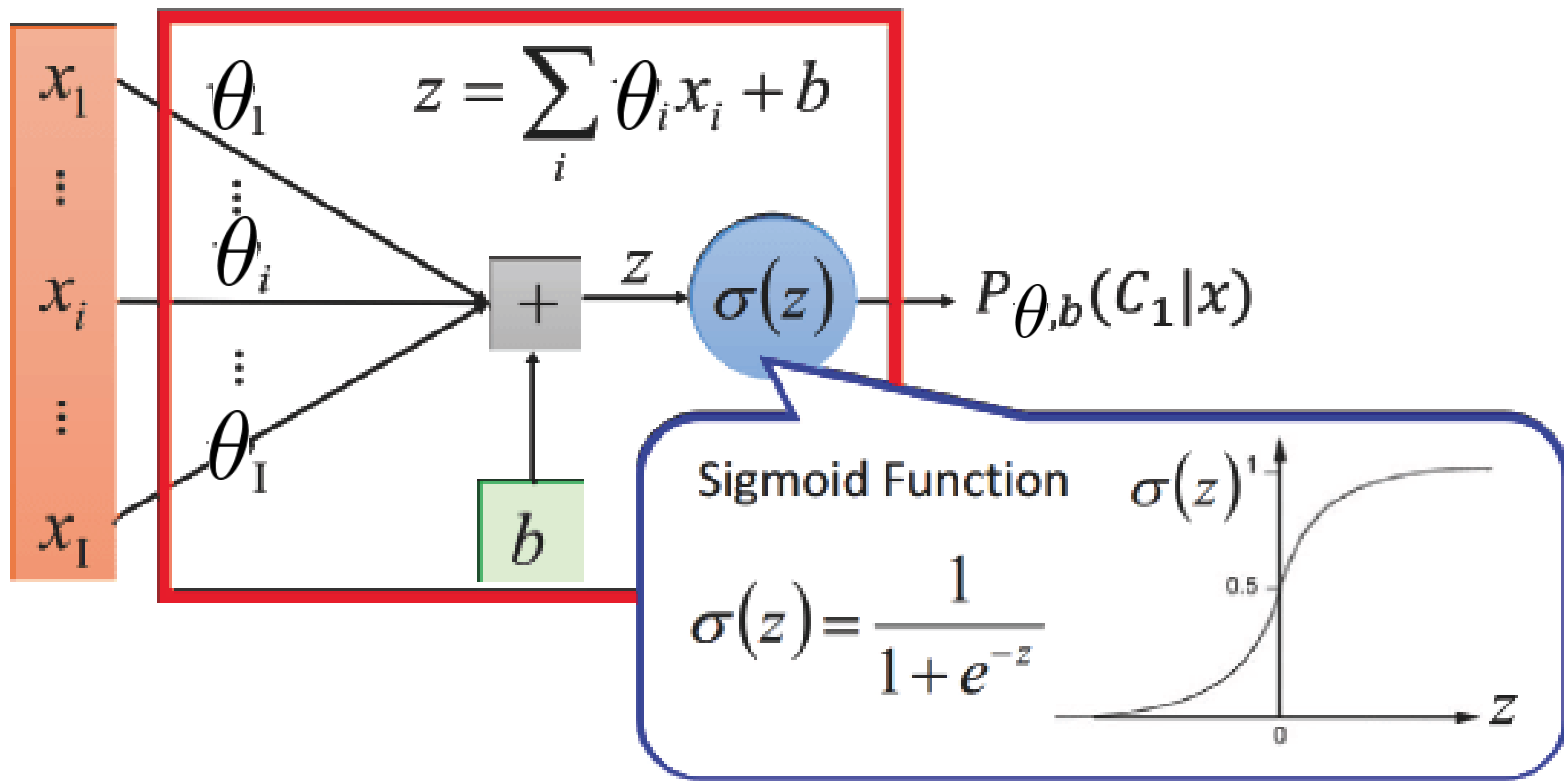
- **Simplest** case (two classes): $y \in \{0, 1\}$

- **Logistic regression model:**

Logistic function

$$p(y = 1|\mathbf{x}, \boldsymbol{\theta}, b) = \frac{1}{1 + \exp(-(\boldsymbol{\theta}^\top \mathbf{x} + b))}$$

$$p(y = 0|\mathbf{x}, \boldsymbol{\theta}, b) = \frac{\exp(-(\boldsymbol{\theta}^\top \mathbf{x} + b))}{1 + \exp(-(\boldsymbol{\theta}^\top \mathbf{x} + b))}$$



For one-dimensional x , $p(y = 1|x, \theta, b) = \frac{1}{1 + \exp(-(\theta^T x + b))}$

Train the Model

- How to find θ and b ? MLE

- Given m labeled samples $(\mathbf{x}^i, y^i)$, $i = 1, \dots, m$
- Find θ and b such that the likelihood of observing the labeled samples is maximized

$$\max_{\theta, b} l(\theta, b) := \log \prod_{i=1}^m P(y^i | \mathbf{x}^i, \theta, b) = \sum_{i=1}^m \log P(y^i | \mathbf{x}^i, \theta, b)$$

- Usually, we equivalently maximize the averaged likelihood

$$\max_{\theta, b} \frac{1}{m} l(\theta, b) := \frac{1}{m} \sum_{i=1}^m \log P(y^i | \mathbf{x}^i, \theta, b)$$

Good news: $l(\boldsymbol{\theta}, b)$ is concave in $(\boldsymbol{\theta}, b)$

a single global optimum

- $P(y = 1|\mathbf{x}, \boldsymbol{\theta}, b) = \frac{1}{1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x} - b)}$
- $P(y = 0|\mathbf{x}, \boldsymbol{\theta}, b) = \frac{\exp(-\boldsymbol{\theta}^\top \mathbf{x} - b)}{1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x} - b)}$
- $\log P(y^i|\mathbf{x}^i, \boldsymbol{\theta}, b) = (y^i - 1)(\boldsymbol{\theta}^\top \mathbf{x}^i + b) - \log(1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x}^i - b))$
- Prove that $-\log(1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x} - b))$ is concave or $\log(1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x} - b))$ is convex.
- If the above holds, $l(\boldsymbol{\theta}, b) := \sum_{i=1}^m \log P(y^i|\mathbf{x}^i, \boldsymbol{\theta}, b)$ is concave

Good news: $l(\boldsymbol{\theta}, b)$ is concave in $(\boldsymbol{\theta}, b)$

a single global optimum

- $P(y = 1|\mathbf{x}, \boldsymbol{\theta}, b) = \frac{1}{1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x} - b)}$
- $P(y = 0|\mathbf{x}, \boldsymbol{\theta}, b) = \frac{\exp(-\boldsymbol{\theta}^\top \mathbf{x} - b)}{1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x} - b)}$
- $\log P(y^i|\mathbf{x}^i, \boldsymbol{\theta}, b) = (y^i - 1)(\boldsymbol{\theta}^\top \mathbf{x}^i + b) - \log(1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x}^i - b))$
- Prove that $-\log(1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x} - b))$ is concave or $\log(1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x} - b))$ is convex. If $f_1, f_2, \dots, f_n$ are convex functions, then $f_1 + f_2 + \dots + f_n$ is also convex
- If the above holds, $l(\boldsymbol{\theta}, b) := \sum_{i=1}^m \log P(y^i|\mathbf{x}^i, \boldsymbol{\theta}, b)$ is concave

Proof: $\log (1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x} - b))$ is convex in $(\boldsymbol{\theta}, b)$

- Consider a point $(\boldsymbol{\theta}_0, b_0)$ and a direction vector $\mathbf{e} = (\boldsymbol{\theta}_1, b_1)$

- $$\begin{aligned} h(t) &:= \log \{1 + \exp[(-\boldsymbol{\theta}_0^\top \mathbf{x} - b_0) + t(-\boldsymbol{\theta}_1^\top \mathbf{x} - b_1)]\} \\ &= \log \{1 + \exp[C_1 + tC_2]\} \end{aligned}$$

- $$h''(t) = \frac{c_2^2}{(1 + \exp[C_1 + tC_2])^2} \exp[C_1 + tC_2] \geq 0$$

- $$h''(0) \geq 0$$

Bad news: no closed form solution to the problem

$$l(\boldsymbol{\theta}, b) := \sum_{i=1}^m (y^i - 1)(\boldsymbol{\theta}^\top \mathbf{x}^i + b) - \log(1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x}^i - b))$$

- Gradient (θ is one-dimension)

$$\frac{\partial l(\theta, b)}{\partial \theta} = \sum_i \left((y^i - 1) x^i + \frac{\exp(-\theta x^i - b) x^i}{1 + \exp(-\theta x^i - b)} \right)$$

$$\frac{\partial l(\theta, b)}{\partial b} = \sum_i \left((y^i - 1) + \frac{\exp(-\theta x^i - b)}{1 + \exp(-\theta x^i - b)} \right)$$

- No closed form solution to the above system of equations

We need to use numerical methods to find $(\boldsymbol{\theta}^*, b^*)$ that maximizes $l(\boldsymbol{\theta}, b)$

Gradient Descent

Gradient Descent Method

- Start with an initial point $x^{(0)}$

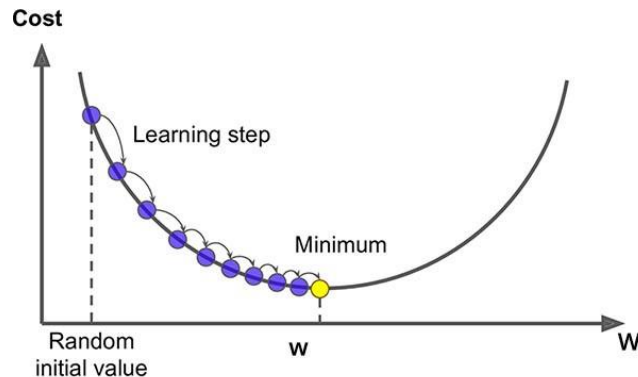
$\alpha^{(t)}$: the step size or learning rate

- Update our point by the following rule (**minimization** problem):

$$x^{(t+1)} = x^{(t)} - \alpha^{(t)} f'(x^{(t)})$$

- Stopping criteria:

- $|x^{(t+1)} - x^{(t)}| \leq \varepsilon$
- or $|f'(x^{(t)})| \leq \varepsilon$



How to select $\alpha^{(t)}$? The selection of $\alpha^{(t)}$ will affect the rate at which we find the local minimizer. A bad selection of $\alpha^{(t)}$ can result in the failure of the algorithm.



Homer descending !

Gradient Descent Method

- Start with an initial point $x^{(0)}$

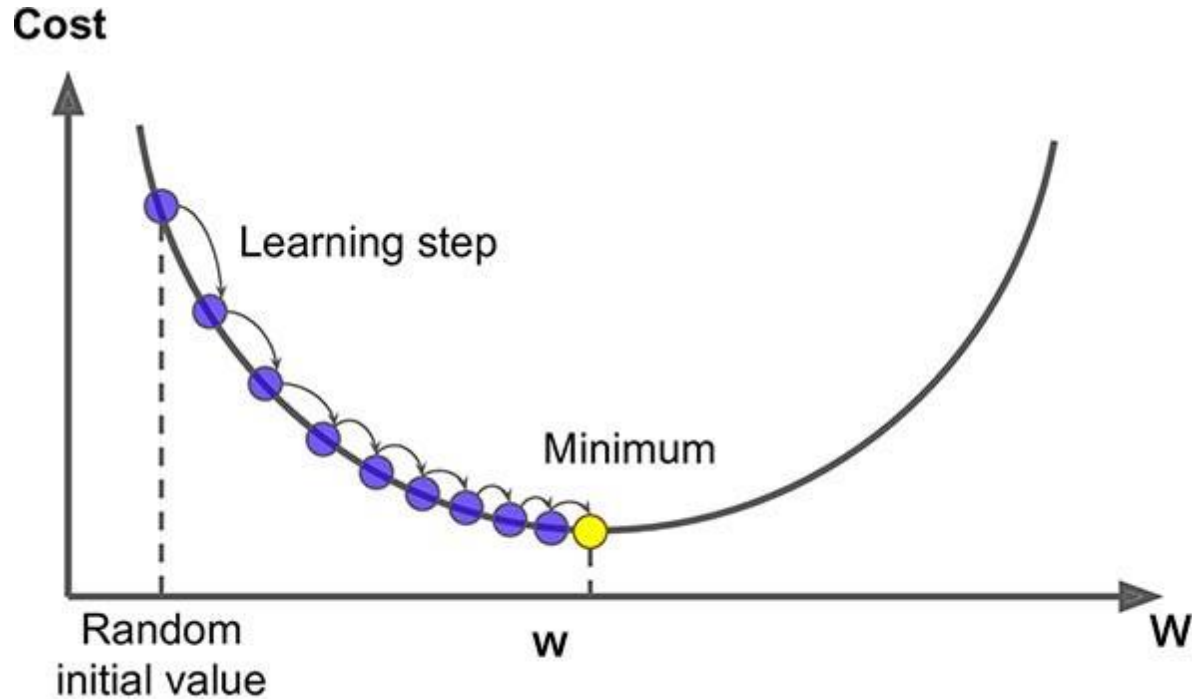
$\alpha^{(t)}$: the step size or learning rate

- Update our point by the following rule (**maximization** problem):

$$x^{(t+1)} = x^{(t)} + \alpha^{(t)} f'(x^{(t)})$$

- Stopping criteria:
 - $|x^{(t+1)} - x^{(t)}| \leq \varepsilon$
 - or $|f'(x^{(t)})| \leq \varepsilon$

We may want the step size, $\alpha^{(t)}$, to be large during the initial steps and smaller as we approach the local minimizer



If $\alpha^{(t)}$ is a big constant value, we may not find meet the stop criteria.

$$f(x) = x^2$$

Suppose $\alpha^{(t)} = 1$.

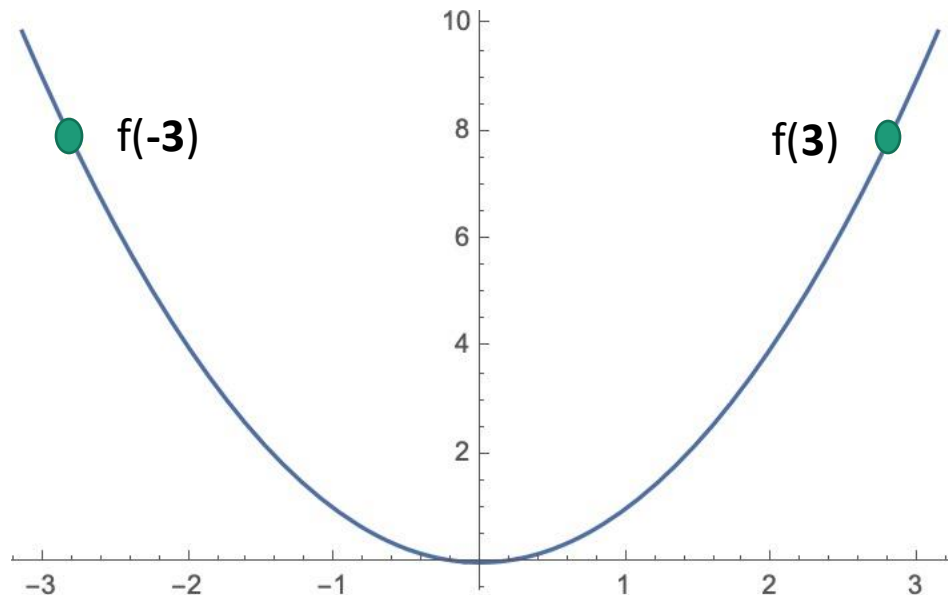
$$x^{(t+1)} = x^{(t)} - f'(x^{(t)})$$

If $x^{(t)} = -3$

- $f'(-3) = -6$
- $x^{(t+1)} = 3$

If $x^{(t)} = 3$

- $f'(3) = 6$
- $x^{(t+1)} = -3$



The updated points will oscillate between 3 and -3

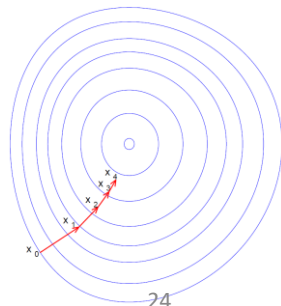
Gradient Descent algorithm for logistic regression

- Initialize parameter $(\boldsymbol{\theta}^0, b^0)$
- While $|\theta^{t+1} - \theta^t| > \epsilon$ or $|b^{t+1} - b^t| > \epsilon$, Do

$$\boldsymbol{\theta}^{t+1} \leftarrow \boldsymbol{\theta}^t + \alpha^{(t)} \frac{1}{m} \sum_i (y^i - 1) \mathbf{x}^i + \frac{\exp(-\boldsymbol{\theta}^{tT} \mathbf{x}^i - b^t) \mathbf{x}^i}{1 + \exp(-\boldsymbol{\theta}^{tT} \mathbf{x}^i - b^t)}$$

$$b^{t+1} \leftarrow b^t + \alpha^{(t)} \frac{1}{m} \sum_i (y^i - 1) + \frac{\exp(-\boldsymbol{\theta}^{tT} \mathbf{x}^i - b^t)}{1 + \exp(-\boldsymbol{\theta}^{tT} \mathbf{x}^i - b^t)}$$

(to maximize the log likelihood)



Gradient Descent algorithm for logistic regression

- If we have a huge amount of data, i.e., m is huge, how to speed up the gradient descent algorithm?
- Use the so-called **stochastic gradient descent** algorithm. See the next page of slide. The main idea is to use a **random subset of data** to **approximate** the gradient direction.

A variant: Stochastic gradient descent

- At each iteration, we randomly choose a small batch of samples in the training data set, and update using the stochastic gradient

$$\begin{aligned}\boldsymbol{\theta}^{t+1} &\leftarrow \boldsymbol{\theta}^t + \alpha^{(t)} \frac{1}{|B|} \sum_{i \in B} (y^i - 1) \mathbf{x}^i + \frac{\exp(-\boldsymbol{\theta}^{tT} \mathbf{x}^i - b^t) \mathbf{x}^i}{1 + \exp(-\boldsymbol{\theta}^{tT} \mathbf{x}^i - b^t)} \\ b^{t+1} &\leftarrow b^t + \alpha^{(t)} \frac{1}{|B|} \sum_{i \in B} (y^i - 1) + \frac{\exp(-\boldsymbol{\theta}^{tT} \mathbf{x}^i - b^t)}{1 + \exp(-\boldsymbol{\theta}^{tT} \mathbf{x}^i - b^t)}\end{aligned}$$

- B : the batch we use in each iteration

Summary

- KNN:
 - Use K nearest neighbors to determine the label. No model parameters
 - We do not need to use all samples in terms of label prediction.
- Logistic regression:
 1. Propose a logistic model (conditional probability)
 2. Use all samples to estimate the parameters
 3. Use the estimated model to calculate the conditional probability for new data
 4. Numerical methods to find the solution (Gradient descent)